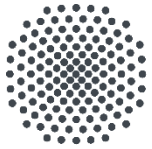
# CMOS Analog Integrated Circuit Design

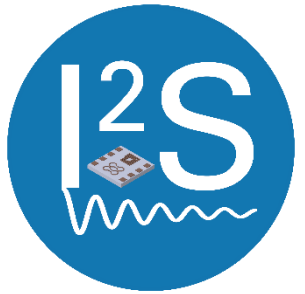
# Chapter 3 – Biasing & Current Sources

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Institute of Smart Sensors
Department of Electrical Engineering & Computer Science
University of Stuttgart

Winter Term 2025/26



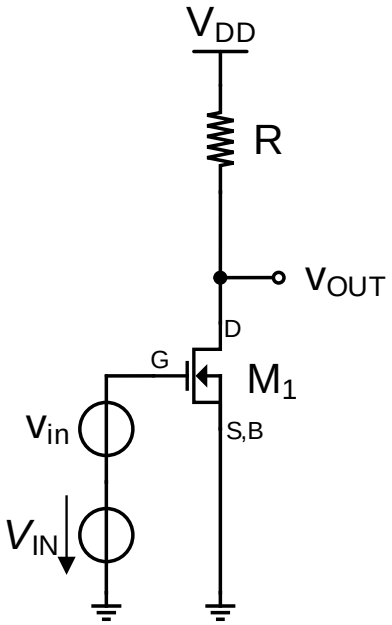
University of Stuttgart
Institute of Smart Sensors



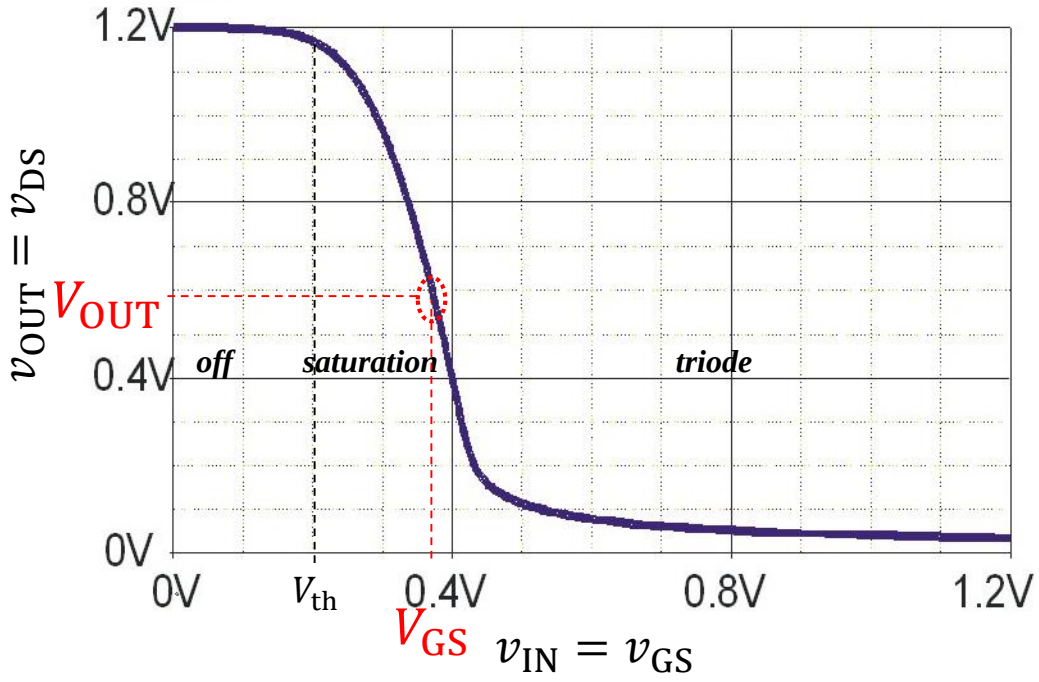
# Overview

- **Amplification – Quick Review**
- Motivation Biasing
- Biasing – Current Mirror
- Cascode Current Source/Sink

# Large-signal Behavior & Bias Point




inverter
with resistive load




$0.12\ \mu\text{m}$ CMOS

$W/L = 10/10$
$V_{DD} = 1.2\ \text{V}$
$R = 100\ \text{k}\Omega$

operating / bias point
$V_{OUT}(V_{IN} = V_{GS})$
for analog operation
(in saturation)

$$v_{IN} = v_{GS} = V_{IN} + v_{in}$$

bias point defined by:

$$V_{OUT} = V_{DS} = V_{DD} - RI_D$$

with $I_D = \frac{\beta}{2}(V_{IN} - V_{th})^2$

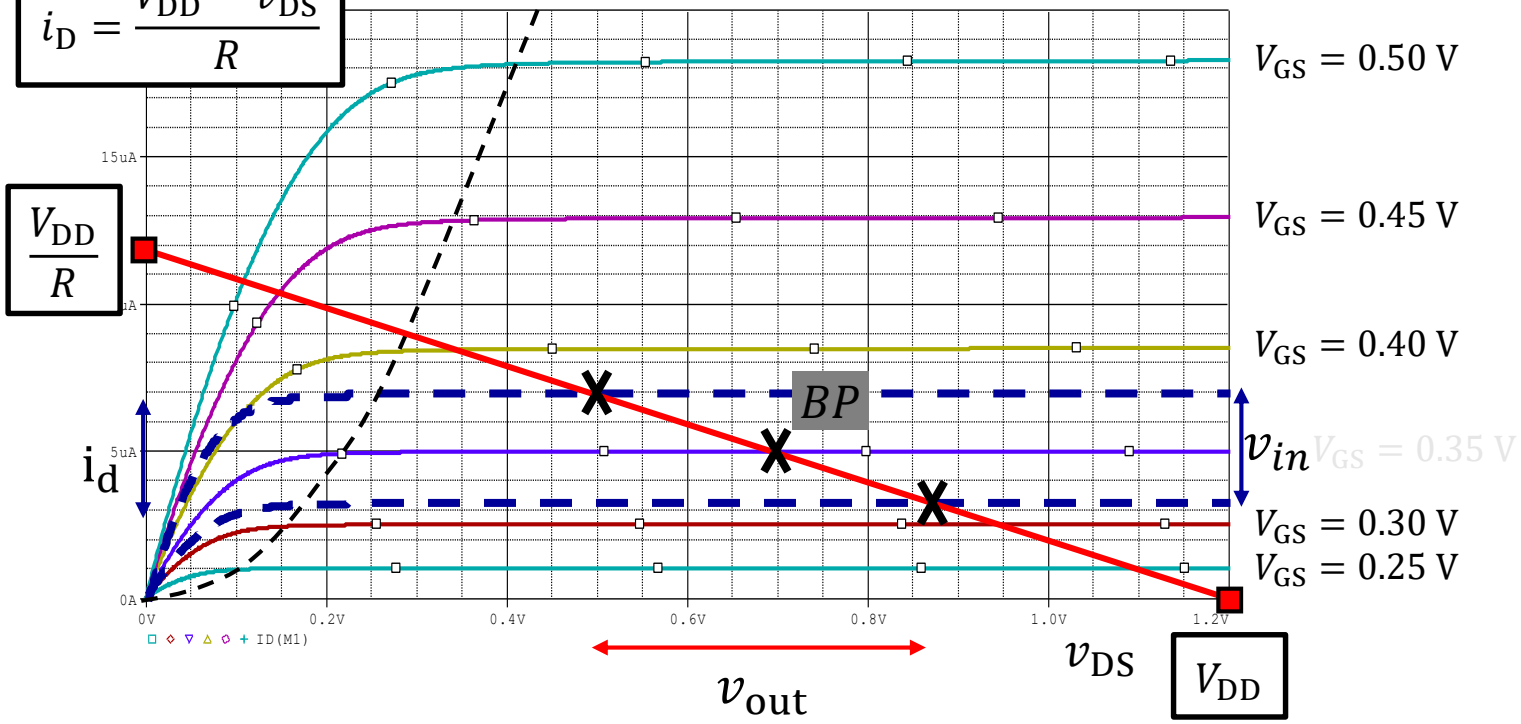
Inverter as an analog amplifier:

Define $I_D$ and thus $v_{OUT}$ by setting $V_{IN}$ for

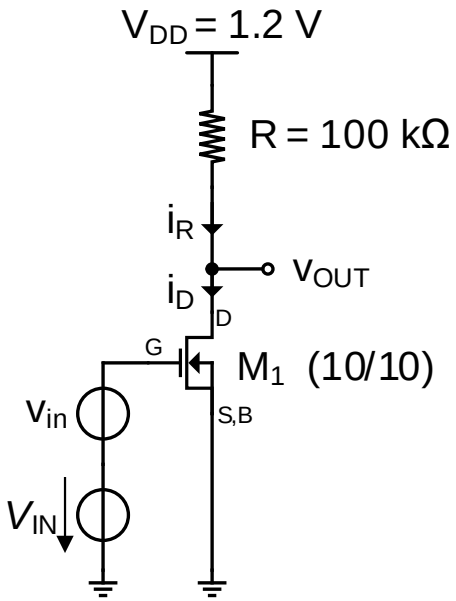
- high amplification of small signal variations $v_{in}$ around $V_{IN}$
- high signal swing $v_{out}$ around $V_{OUT}$

# Bias Point & Graphical Illustration of Amplification

$$i_D = \frac{V_{DD} - v_{DS}}{R}$$




$$i_D = \frac{\beta_n}{2}(v_{IN} - V_{th})^2$$
$$v_{OUT} = V_{DD} - Ri_D$$
$$i_D = i_R$$
$$i_d = G_m v_{in}$$
$$v_{out} = -Ri_d = -G_m R v_{in}$$




$$v_{in} = v_{gs} \qquad v_{in} \longrightarrow i_d$$
$$v_{out} = v_{ds} \qquad i_d \longrightarrow v_{out}$$

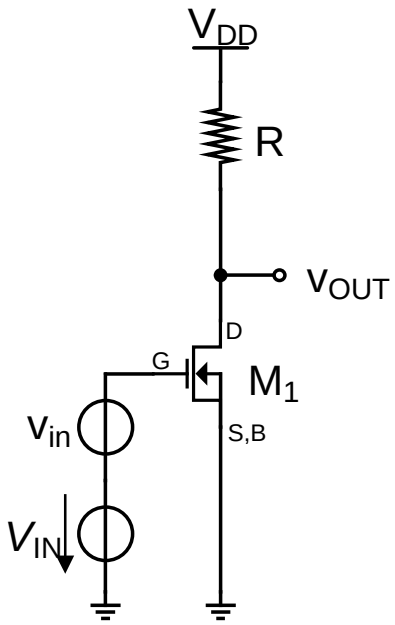
$$A = \frac{v_{out}}{v_{in}}$$

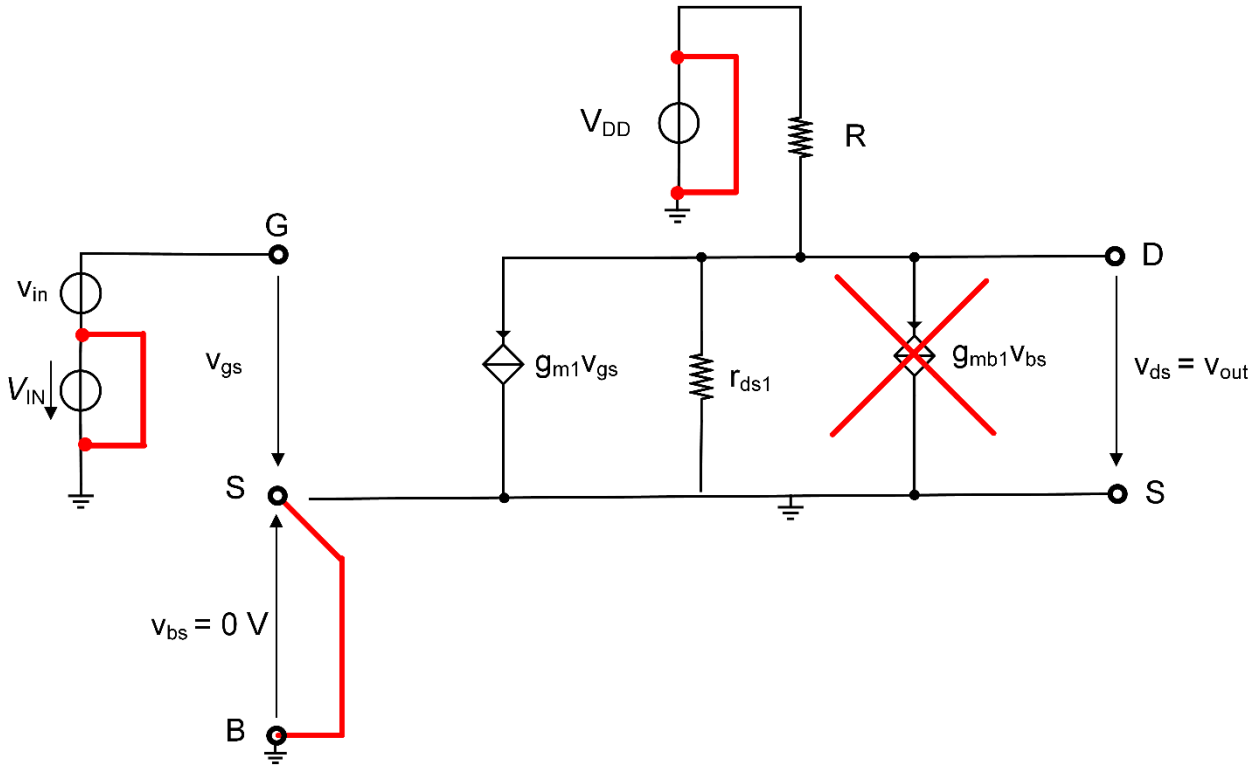
transconductance

$$G_m = \frac{i_{out}}{v_{in}} = \frac{i_d}{v_{gs}}$$

$$A = \frac{v_{out}}{v_{in}} = -G_m R$$

# Amplification – Small-signal Equivalent Circuit



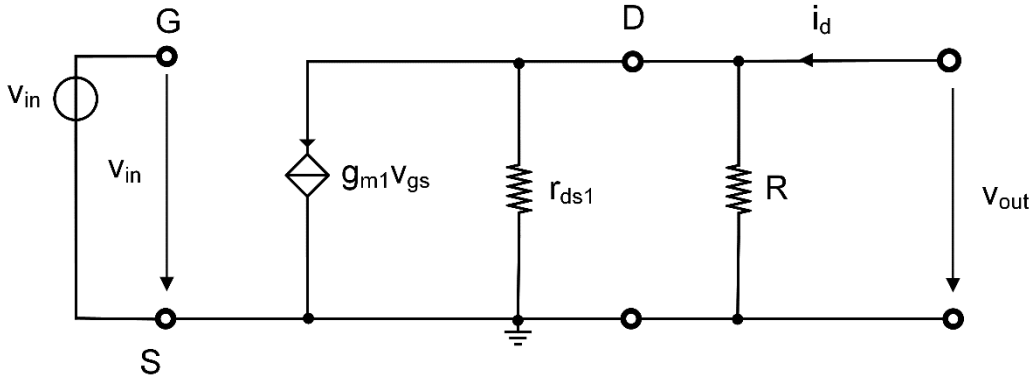




Replace all <u>constant</u> voltage-sources by short-circuits!

$(R_{\mathrm{IN}} = 0\ \Omega)$

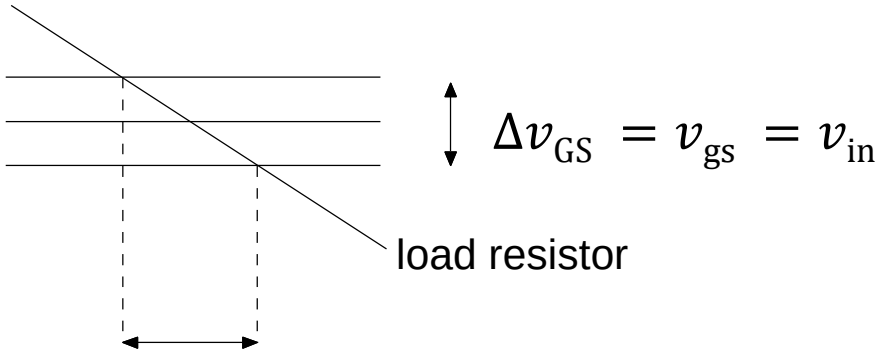
Replace all <u>constant</u> current-sources by open-circuits!

$(R_{\mathrm{IN}} = \infty\ \Omega)$



# MOS Transistor – Small-signal Output Resistance

## no channel-length modulation




$\Delta v_{\mathrm{GS}} = v_{\mathrm{gs}} = v_{\mathrm{in}}$

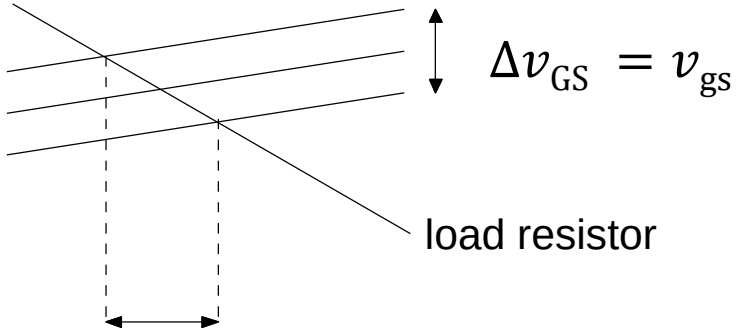
$\Delta v_{\mathrm{DS}} = v_{\mathrm{ds}} = v_{\mathrm{out}}$

$$i_{\mathrm{D}} = \frac{\beta}{2}(v_{\mathrm{GS}} - V_{\mathrm{th}})^2$$

$$\left.\frac{di_{\mathrm{D}}}{dv_{\mathrm{DS}}}\right|_{BP} = 0\ \mathrm{S} \Rightarrow g_{\mathrm{ds}} = 0\ \mathrm{S}$$

$$\boxed{\Rightarrow r_{\mathrm{ds}} = \infty\ \Omega}\ \text{for } \lambda \ll 0$$

## channel-length modulation




$\Delta v_{\mathrm{GS}} = v_{\mathrm{gs}}$

$\Delta v_{\mathrm{DS}} = v_{\mathrm{ds}} = v_{\mathrm{out}}$

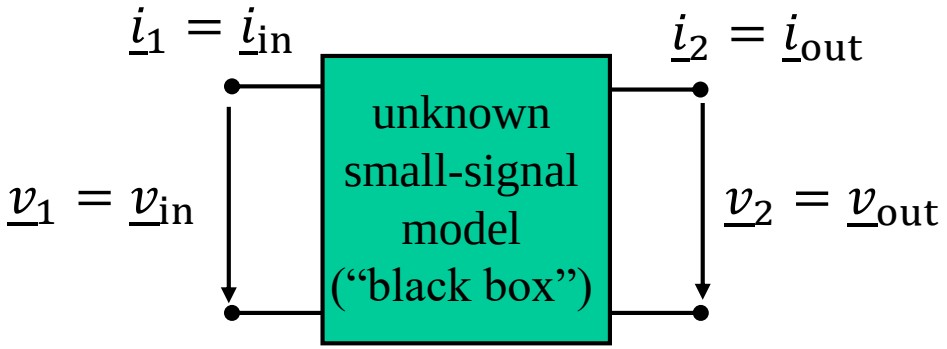
$$i_{\mathrm{D}} = \frac{\beta}{2}(v_{\mathrm{GS}} - V_{\mathrm{th}})^2(1 + \lambda v_{\mathrm{DS}})$$

$$\left.\frac{di_{\mathrm{D}}}{dv_{\mathrm{DS}}}\right|_{BP} \approx \lambda I_{\mathrm{D}} \qquad \boxed{\Rightarrow \frac{1}{r_{\mathrm{ds}}} = g_{\mathrm{ds}} = \lambda I_{\mathrm{D}}}$$

for $\lambda \approx 0.01 \ldots 0.05\ \mathrm{V}^{-1}$
@ $\mathrm{L} = 10\ \mu\mathrm{m}$

# Determination Small-signal Amplification – Approach I

Derive the small-signal amplification from the small-signal model in two steps! (beginner)



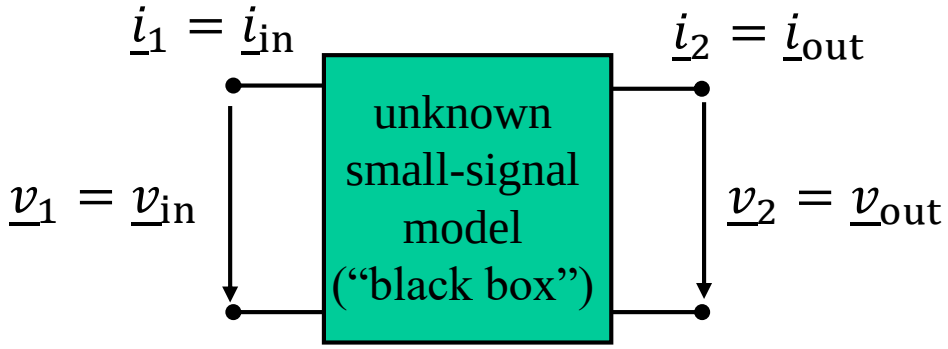

The "black box" is characterized by the characteristics of the currents and voltages at the input & output
→ complex admittance parameters

$$\underline{y}_{11}, \underline{y}_{12}, \underline{y}_{21}, \underline{y}_{22}$$

complex admittance $\underline{Y} = \frac{1}{\underline{Z}}$ ↔ complex impedance

# Determination Small-signal Amplification – Approach I

Derive the small-signal amplification from the small-signal model in two steps!
(beginner)




$$(G_m =) y_{21} = \left.\frac{i_{out}}{v_{in}}\right|_{v_{out} = 0\ V}$$

$$(g_{out} =) y_{22} = \left.\frac{i_{out}}{v_{out}}\right|_{v_{in} = 0\ V}$$

input admittance ~~reverse transfer admittance~~

$$i_{in} = y_{11} v_{in} + y_{12} v_{out}$$

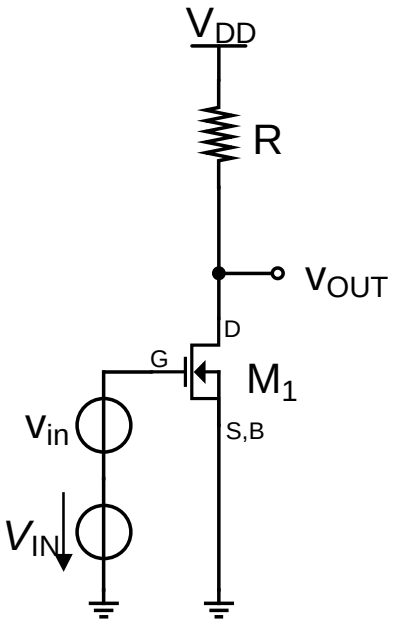
$$i_{out} = y_{21} v_{in} + y_{22} v_{out}$$

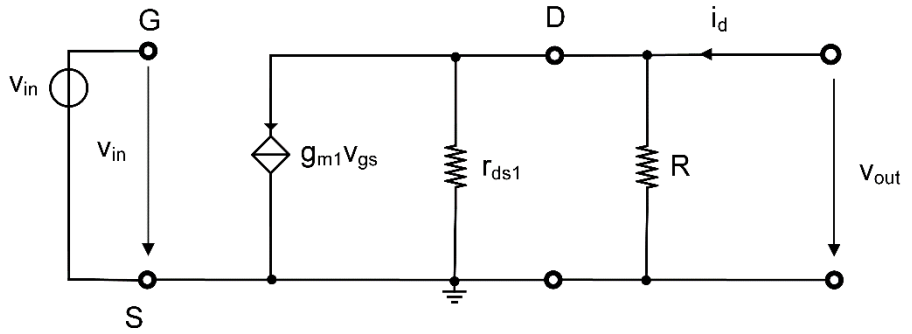
transconductance

output admittance

$$A = -G_m r_{out} = -\frac{G_m}{g_{out}} = -\frac{y_{21}}{y_{22}}$$

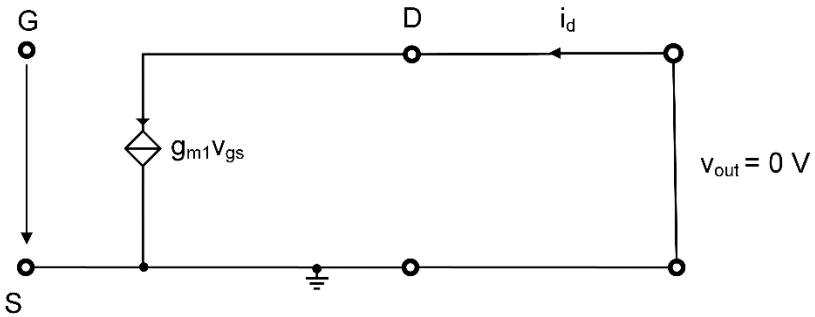
# Approach I – Determine $G_m$ & $r_{out}$




Why do we not "see" $r_{ds1}$ in the graphical discussion?

short-circuit current

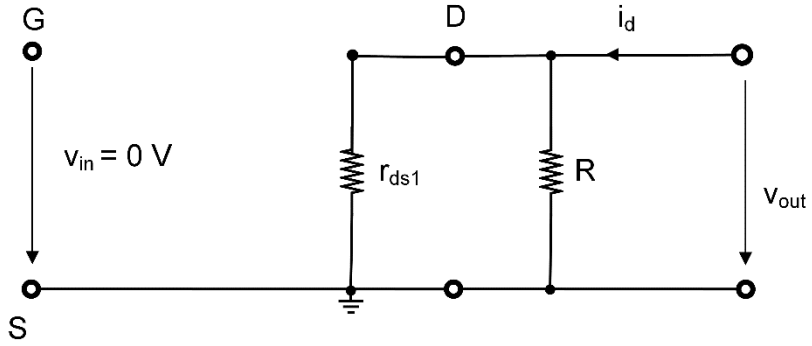



$$G_m = \left.\frac{i_d}{v_{in}}\right|_{v_{out}=0\,V}$$

$$i_d = g_{m1} v_{in}$$

$$G_m = g_{m1}$$

open-circuit voltage




$$r_{out} = \left.\frac{v_{out}}{i_d}\right|_{v_{in}=0\,V}$$

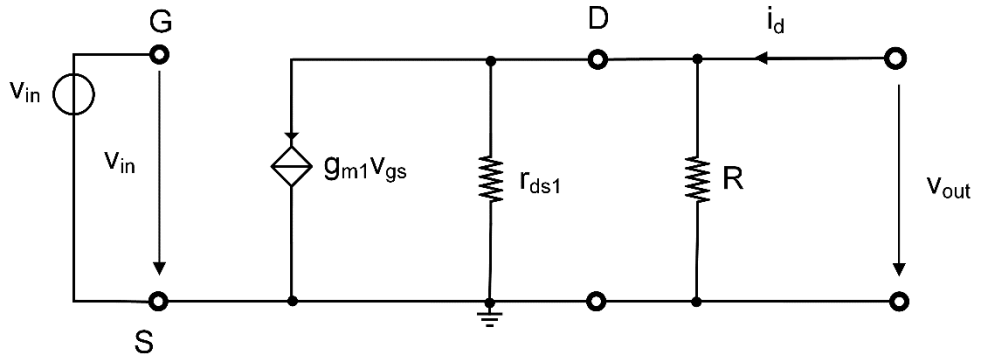
$$v_{out} = i_d (R \,||\, r_{ds1})$$

$$r_{out} = (R \,||\, r_{ds1}) = \frac{R\, r_{ds1}}{R + r_{ds1}}$$

$$A = -g_{m1} r_{out}$$

# Determination Small-signal Amplification – Approach II

Derive the small-signal amplification straightforward
from the small-signal model!
(experienced designer)



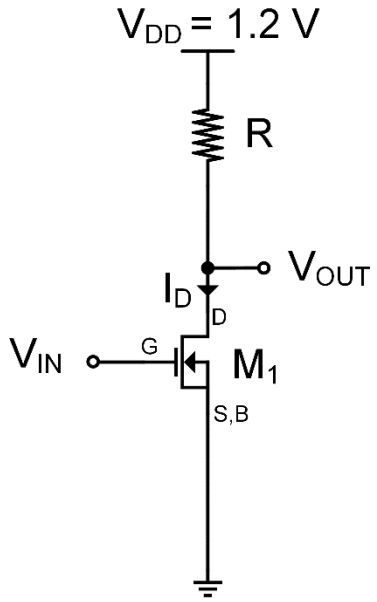

$$A = -g_{\mathrm{m1}}\, r_{\mathrm{ds1}} || R$$

Why do we not “see” $r_{\mathrm{ds1}}$
in the graphical discussion?

# Overview

- Amplification – Quick Review
- **Motivation Biasing**
- Biasing – Current Mirror
- Cascode Current Source/Sink

# Resistive Load – Example




process parameters

$$\beta_0 = 370\ \frac{\mu A}{V^2}$$

$$\lambda = 0.02\ \frac{1}{V}$$

$$V_{th} = 0.19\ V$$

operating point

$$V_{IN} = 0.35\ V$$

$$V_{OUT} = 0.7\ V$$

$$V_{SB} = 0\ V$$

dimensioning

$$R = 100\ k\Omega$$

$$\frac{W}{L} = \frac{10\ \mu m}{10\ \mu m}$$

We thus get

$$I_D = \frac{\beta}{2}(V_{GS} - V_{th})^2 \cdot (1 + \lambda V_{DS}) \Rightarrow I_D \approx 4.8\ \mu A$$

$$g_m = \beta(V_{GS} - V_{th}) \Rightarrow g_{m\,1} \approx 59\ \mu S$$

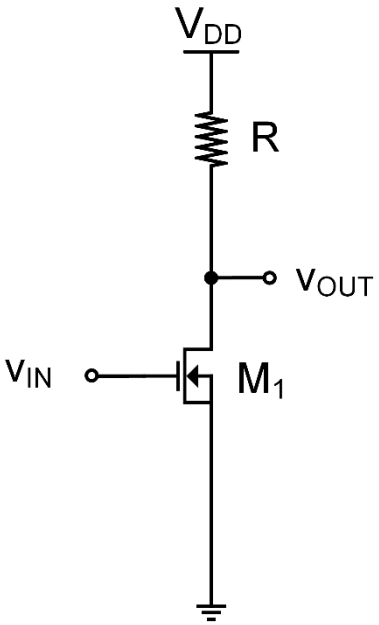
$$g_{DS} = \lambda I_D \Rightarrow g_{DS1} \approx 0.096\ \mu S$$

$$\Rightarrow r_{DS1} \approx 10\ M\Omega$$

$$A = -g_{m1} r_{out} \approx 6$$

$$\boldsymbol{R \uparrow \rightarrow V_{OUT} = ?}$$

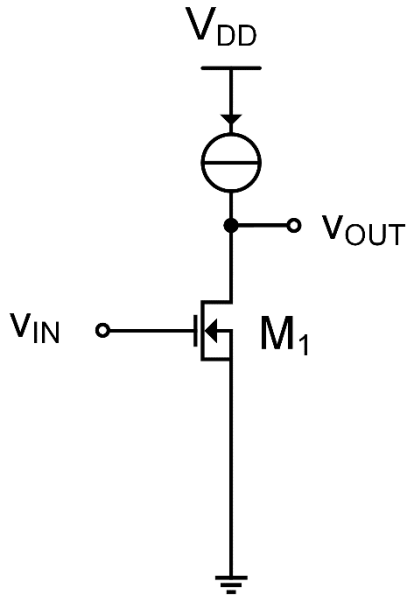
# Biasing – Motivation




$$A_V = -g_\mathrm{m}(R \parallel r_\mathrm{ds})$$

Resistor *R* defines the gain of the inverting amplifier and needs to be large.

But it also defines the bias point of the output and can thus not be very large for reasonable currents.




$$A_V = -g_\mathrm{m} r_\mathrm{ds}$$

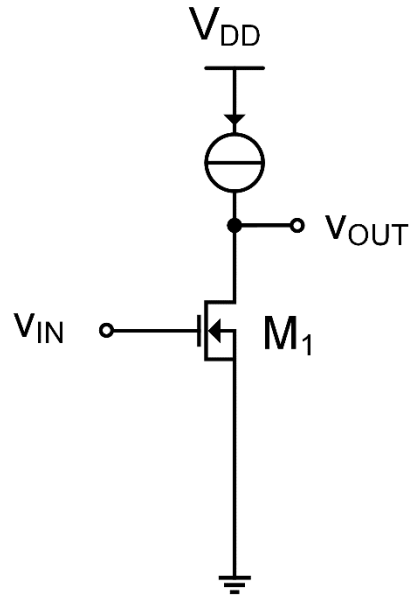
The load element needs to have a low *large-signal* resistance but a high *small-signal* resistance!

Amplifier should ideally be biased by a current source.

# Intrinsic Gain of a MOS Transistor

For any arbitrary load resistance $r \gg r_{ds}$

$$A_V \approx -g_m r_{ds} = -\frac{g_m}{g_{ds}} = -59\ \mu S\ 10\ M\Omega = -590$$ intrinsic transistor gain




How can we increase the intrinsic transistor gain?

$$g_m = \beta_0 \frac{W}{L}(V_{GS} - V_{th}) \qquad g_m \propto \frac{W}{L}$$

$$g_{ds} = \lambda I_D \qquad \lambda \propto \frac{1}{L} \quad I_D \propto \frac{W}{L}$$

$$L \uparrow \Rightarrow g_m \downarrow \quad I \downarrow, \lambda \downarrow \Rightarrow g_{DS} \downarrow\downarrow \Rightarrow A_V \uparrow$$

$$\Rightarrow A = -g_m r_{ds} \approx \frac{\beta_0 \frac{W}{L}(V_{GS} - V_{th})}{\frac{\Delta L}{L}\frac{1}{2}\beta_0 \frac{W}{L}(V_{GS} - V_{th})^2} \propto \frac{2L}{\Delta L (V_{GS} - V_{th})}$$

$g_m$ is also responsible for speed!
trade-off:
gain (long L) ↔ speed (short L)

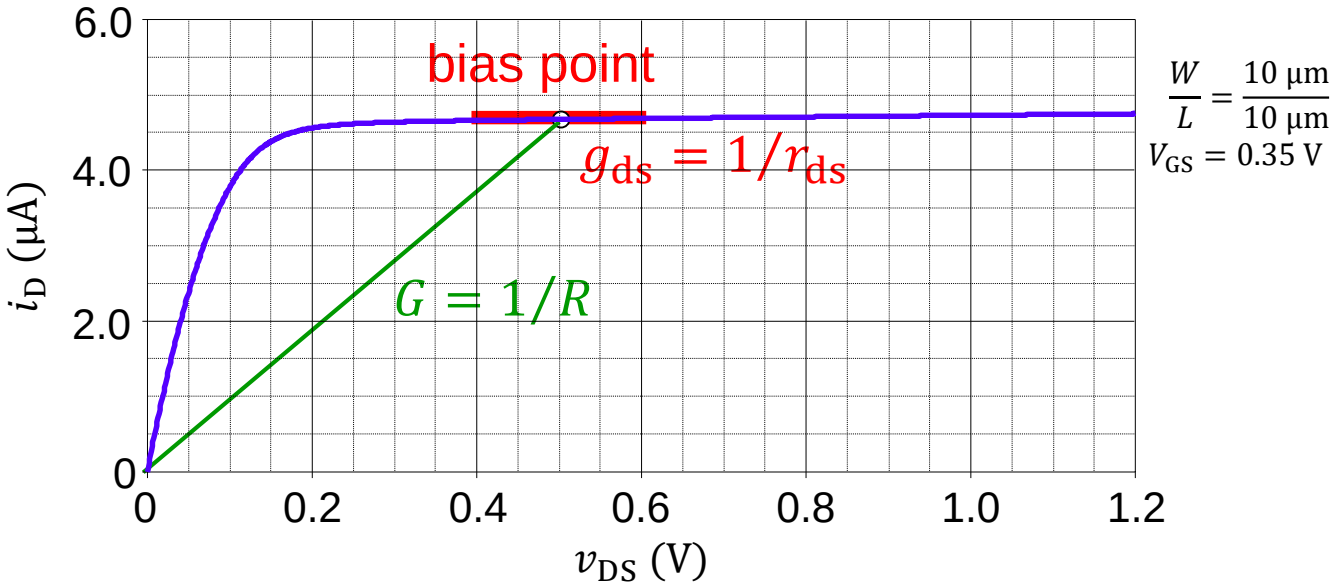
# Motivation – Transistors as „Resistors“

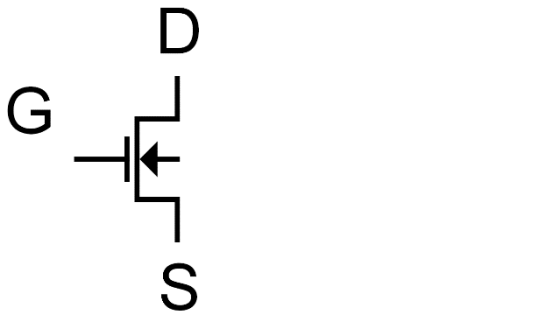
MOS transistors are voltage-controlled current sources.

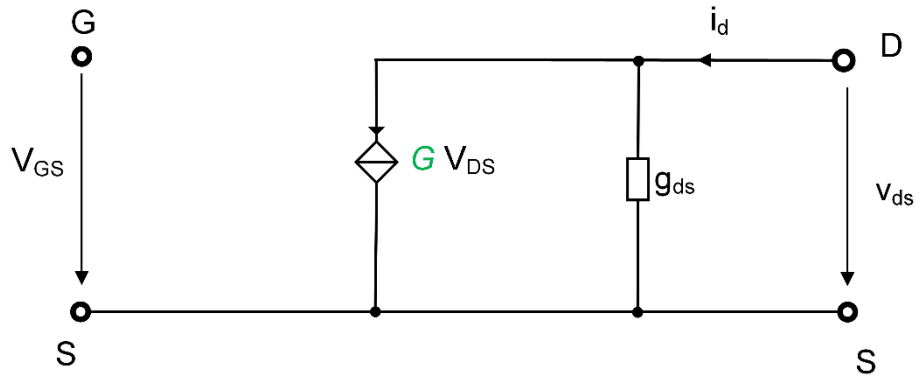
MOS transistors can behave as (small-signal) resistors.

The equivalent resistance depends on the operating point.

example: gate tied to fixed voltage



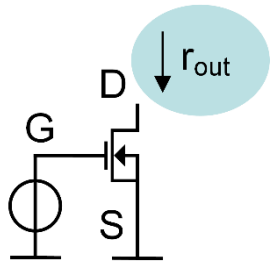

$$G = \frac{1}{R} = \frac{I_D}{V_{DS}}$$

$$\frac{\partial i_D}{\partial v_{DS}} = g_{ds} = \frac{1}{r_{ds}}$$

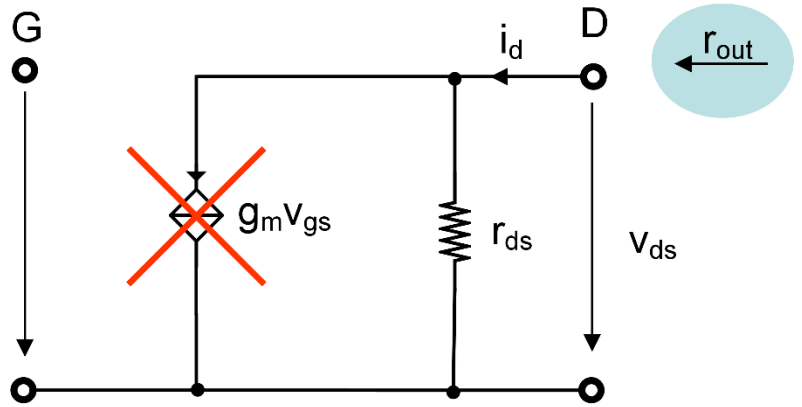
Fixed voltage?

# Transistor with Constant $V_{GS}$

## a) n-MOS



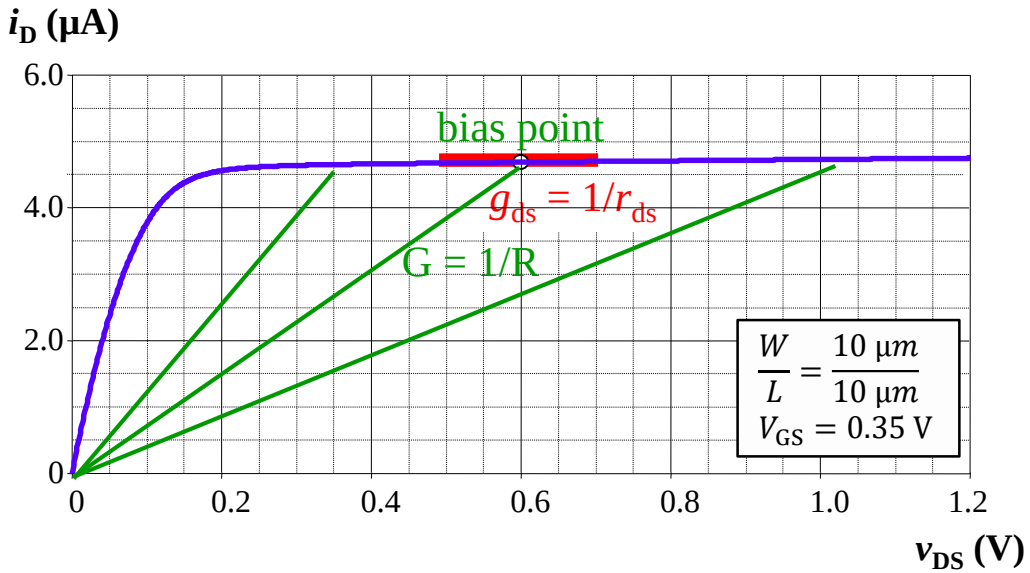

≙



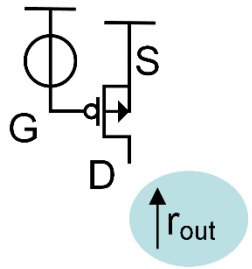

$$r_{out} = \frac{v_{out}}{i_{out}} = \frac{v_{ds}}{i_d}$$

$$\boxed{r_{out} = r_{ds}}$$

$$\frac{1}{r_{ds}} = g_{ds} = \frac{\partial i_D}{\partial v_{DS}} \approx \lambda I_D$$

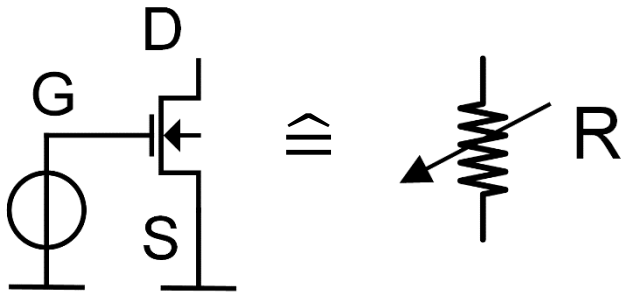



## b) p-MOS



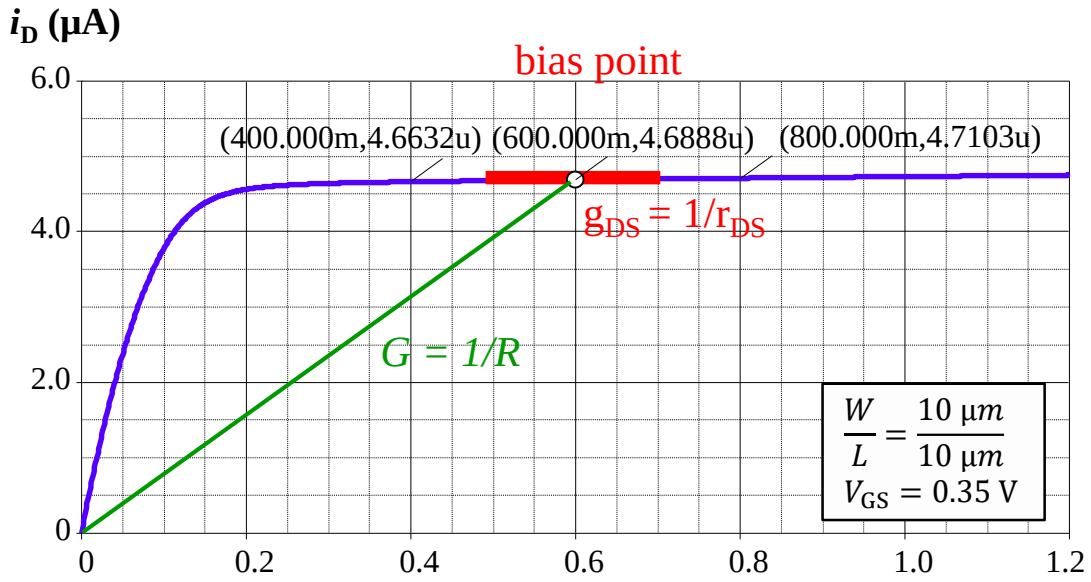

$$\boxed{r_{out} = r_{ds}}$$

# Transistor with Constant $V_{GS}$ – Example




The equivalent output resistance depends on the operating point ($V_{GS}$).

$$r_{out} = r_{ds}$$




small-signal resistance (gain)

$$\frac{\partial i_D}{\partial v_{DS}} = g_{ds} = \frac{1}{r_{ds}} \approx \lambda I_D \approx 0.02\ V^{-1}\ 4.74\ \mu A \approx 0.1\ \mu S$$

$$\Rightarrow r_{ds} \approx 10 M\Omega \quad \text{calculation}$$

$$g_{ds} = \frac{\Delta i_D}{\Delta v_{DS}} = \frac{4.71\ \mu A - 4.66\ \mu A}{800\ mV - 400\ mV} = 0.125\ \mu S$$

$$\Rightarrow r_{ds} = 8\ M\Omega \quad \text{simulation}$$

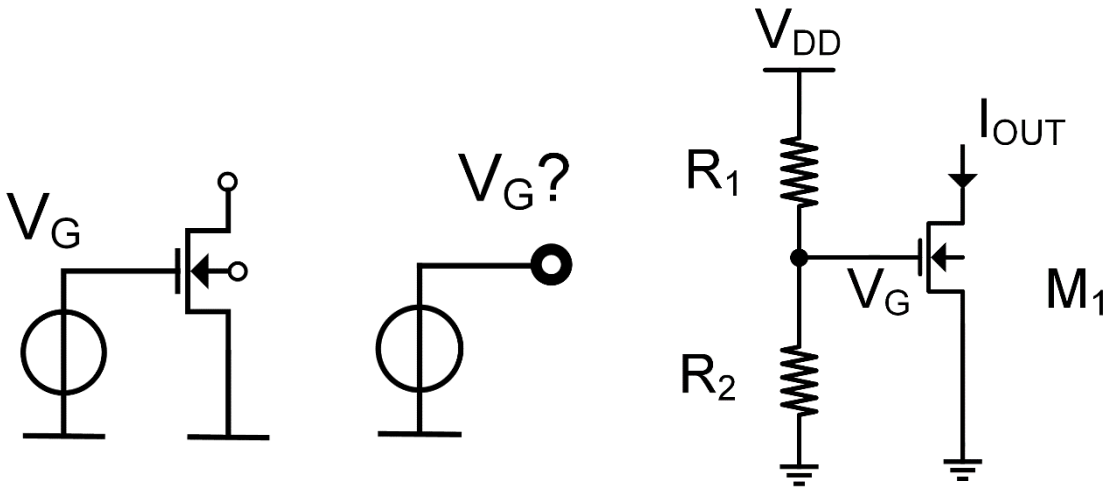
large-signal resistance (bias point)

$$G = \frac{1}{R} = \frac{I_D}{V_{DS}} = \frac{4.69\ \mu A}{600\ mV} \approx 7.8\ \mu S$$

$$\Rightarrow R \approx 128\ k\Omega$$

# Bias Voltage – Generation?

How to generate a constant voltage?




*Can we use transistors instead of resistors for biasing?*

$$I_{\mathrm{OUT}} = \frac{1}{2}\beta_0 \frac{W}{L}(V_{\mathrm{GS}} - V_{\mathrm{th}})^2$$

$$= \frac{1}{2}\mu_{\mathrm{n}} C'_{ox} \frac{W}{L}\left(\frac{R_2}{R_1 + R_2}V_{\mathrm{DD}} - V_{\mathrm{th}}\right)^2$$

$I_{\mathrm{OUT}}$ depends on

- supply voltage $V_{\mathrm{DD}}$
- process parameters $C'_{ox}$, $\mu_{\mathrm{n}}$, $V_{\mathrm{th}}$, $W$, $L$, $R_1$, $R_2$
- temperature $\mu_{\mathrm{n}}$, $V_{\mathrm{th}}$

# Summary

Motivation for biasing:

- Transistors can be used as “resistors” whose large-signal and small-signal resistances are different (*small* large-signal resistance vs. *large* small-signal resistance).
- A transistor operated in saturation with a constant gate-source voltage is an almost “perfect” current source → very high small-signal resistance.
- The generation of a bias voltage can be performed by means of a voltage divider. However, process variations cause voltage variations!
- Many microelectronic circuits are thus biased by current mirrors (next sub-chapter).

# Literature

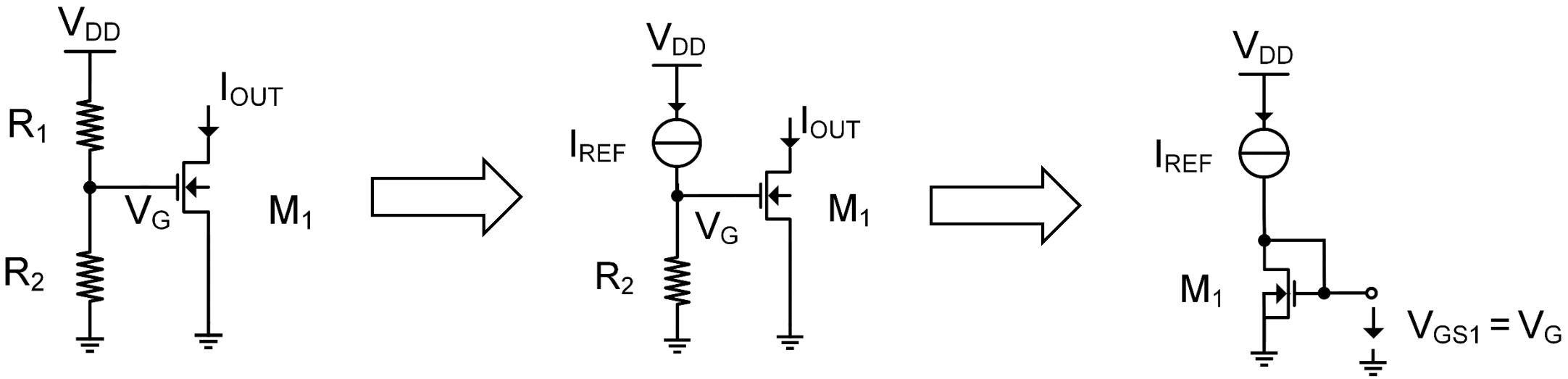
## Literature

- Razavi, chapters 3.2.2 & 3.2.3, page 53ff

# Overview

- Amplification – Quick Review
- Motivation Biasing
- **Biasing – Current Mirror**
  - **Biasing Based on Diodes**
  - Ideal vs. Real Current Mirror
  - Small-signal Equivalent Circuit
- Cascode Current Source/Sink

# Current Mirror



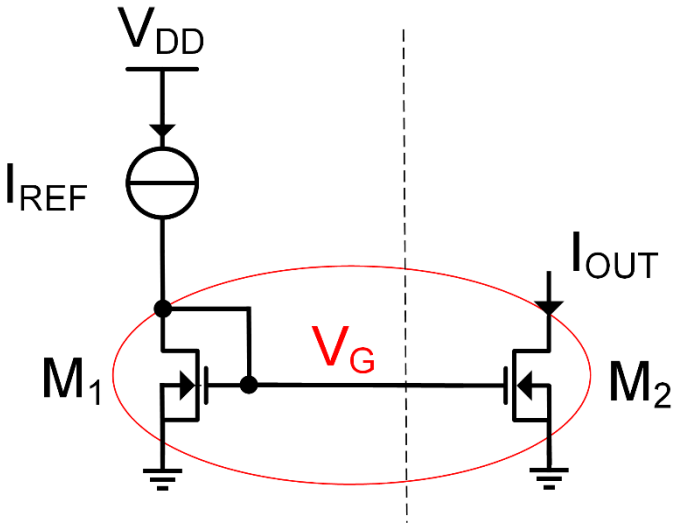

bias circuit

$I_{\mathrm{REF}}$ determines $V_{\mathrm{GS1}} = V_{\mathrm{G}}$

$V_{\mathrm{GS1}} = V_{\mathrm{GS2}}$ determines $I_{\mathrm{OUT}}$

$$\Rightarrow I_{\mathrm{OUT}} = \frac{\beta_0}{2}\frac{W}{L}(V_{\mathrm{GS1}} - V_{\mathrm{th}})^2$$

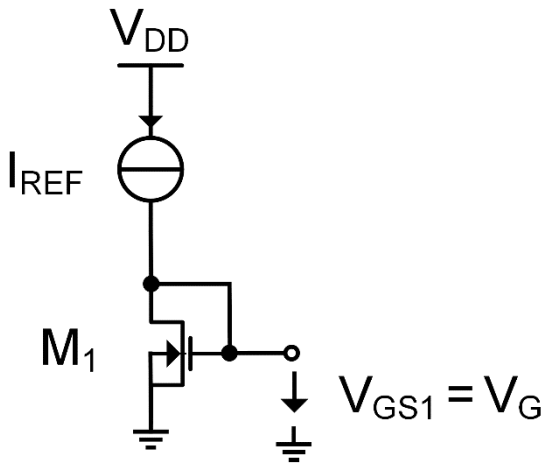
$\Rightarrow I_{\mathrm{REF}}$ is „mirrored“ into $I_{\mathrm{OUT}}$

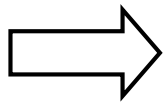


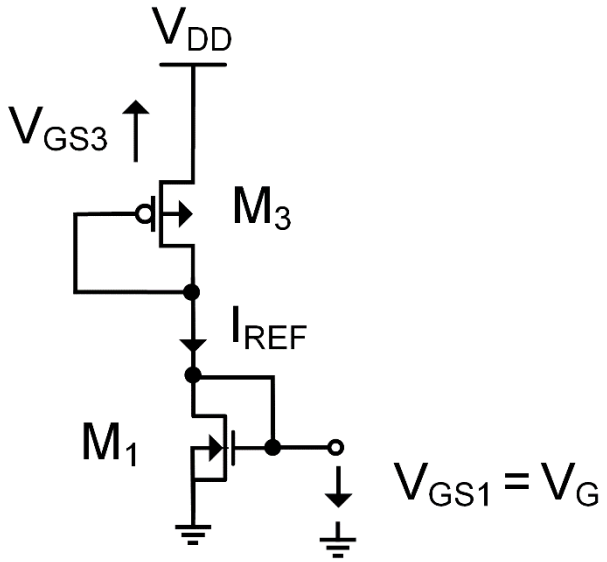

current mirror

# Bias Circuit – Implementation

How to implement $I_{\mathrm{REF}}$?






$$V_{\mathrm{DD}} - V_{\mathrm{SS}} = |V_{\mathrm{GS3}}| + V_{\mathrm{GS1}}$$

If $V_{\mathrm{GS1}} \approx |V_{\mathrm{GS3}}|$

$$V_{\mathrm{GS1}} = \frac{V_{\mathrm{DD}} - V_{\mathrm{SS}}}{2}$$

$$V_{\mathrm{GSeff}} = V_{\mathrm{GS}} - V_{\mathrm{th}}$$

if $V_{\mathrm{th1}} \approx |V_{\mathrm{th3}}|\ (= 0.2\ \mathrm{V})$

$$V_{\mathrm{GSeff1,3}} = \left.\frac{V_{\mathrm{DD}} - V_{\mathrm{SS}} - 2V_{\mathrm{th}}}{2}\right|_{V_{\mathrm{DD}} - V_{\mathrm{SS}} = 1.2\ \mathrm{V}} \approx 400\ \mathrm{mV}$$

# Reducing the Bias Current

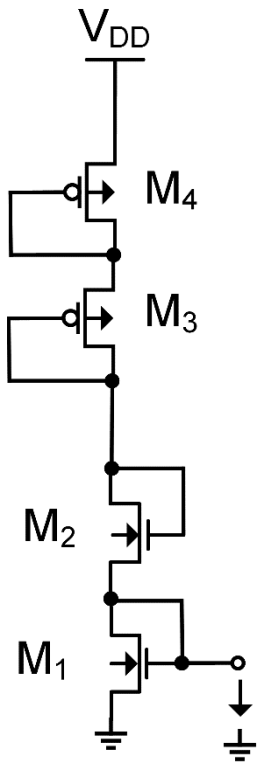
How to reduce the bias current?

Increase the resistance of each transistor or the number of transistor!

bias current: $I_D = \frac{\beta_0}{2}\frac{W}{L}(V_{GSeff})^2$

a) reduce $W/L$ since $I_D \propto \frac{W}{L}$

b) reduce $V_{GSeff}$ since $I_D \propto (V_{GSeff})^2$




$$V_{GSeff} = \left.\frac{V_{DD} - V_{SS} - 4V_{th}}{4}\right|_{V_{DD}+|V_{SS}| = 1.2\ \mathrm{V}} = \frac{1.2\ \mathrm{V} - 0.8\ \mathrm{V}}{4} = 100\ \mathrm{mV}$$

if $V_{th1} \approx |V_{th3}|\ (= 0.2\ \mathrm{V})$

- $V_{GSeff}$ reduced by factor 4
- $I_D$ reduced by factor 16!

Bias current is not well defined
(no reference current source)
but can be copied to many circuit blocks
= bias circuit with many bias voltages.

# Overview

- Amplification – Quick Review
- Motivation Biasing
- **Biasing – Current Mirror**
  - Biasing Based on Diodes
  - **Ideal vs. Real Current Mirror**
  - Small-signal Equivalent Circuit
- Cascode Current Source/Sink

# Current Mirror – Ideal Characteristic

assumptions:

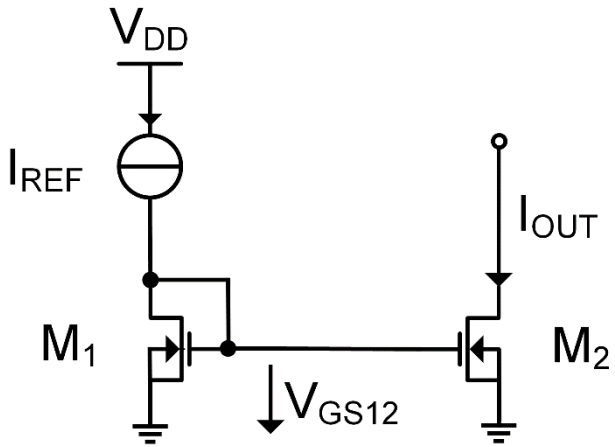
$$V_{GS1} = V_{GS2} = V_{GS12}$$

$$V_{th1} = V_{th2} = V_{th}$$

$$\beta_{01} = \beta_{02} = \beta_0$$

current mirror

bias circuit




current source

$$I_{REF} = I_{D1} = \frac{\beta_0}{2}\frac{W}{L}\bigg|_1 (V_{GS12} - V_{th})^2$$

$$I_{OUT} = I_{D2} = \frac{\beta_0}{2}\frac{W}{L}\bigg|_2 (V_{GS12} - V_{th})^2$$

$$I_{OUT} \propto I_{REF}$$

$$\frac{I_{OUT}}{I_{REF}} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}$$

# Current Mirror – Real Characteristic

$$I_{REF} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_1 (V_{GS12} - V_{th})^2$$

$$I_{OUT} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_2 (V_{GS12} - V_{th})^2$$

$$\frac{I_{OUT}}{I_{REF}} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}$$

If $M_1$ and $M_2$ are equal, then $I_{D1} = I_{D2}$.

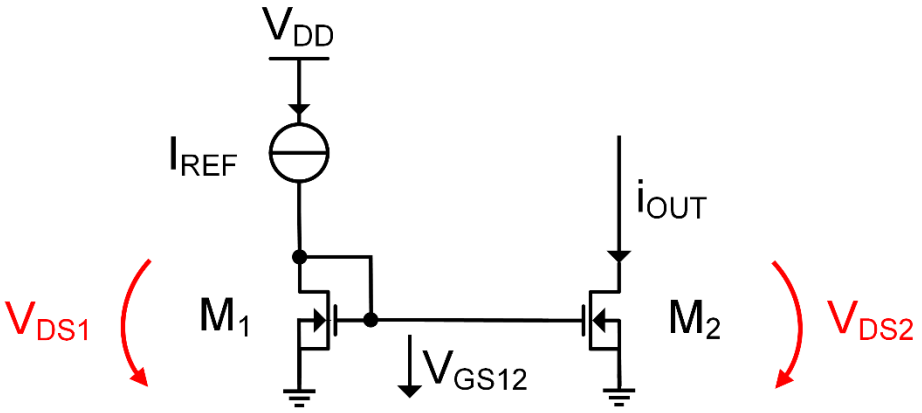
However, only if $V_{DS1} = V_{DS2}$!

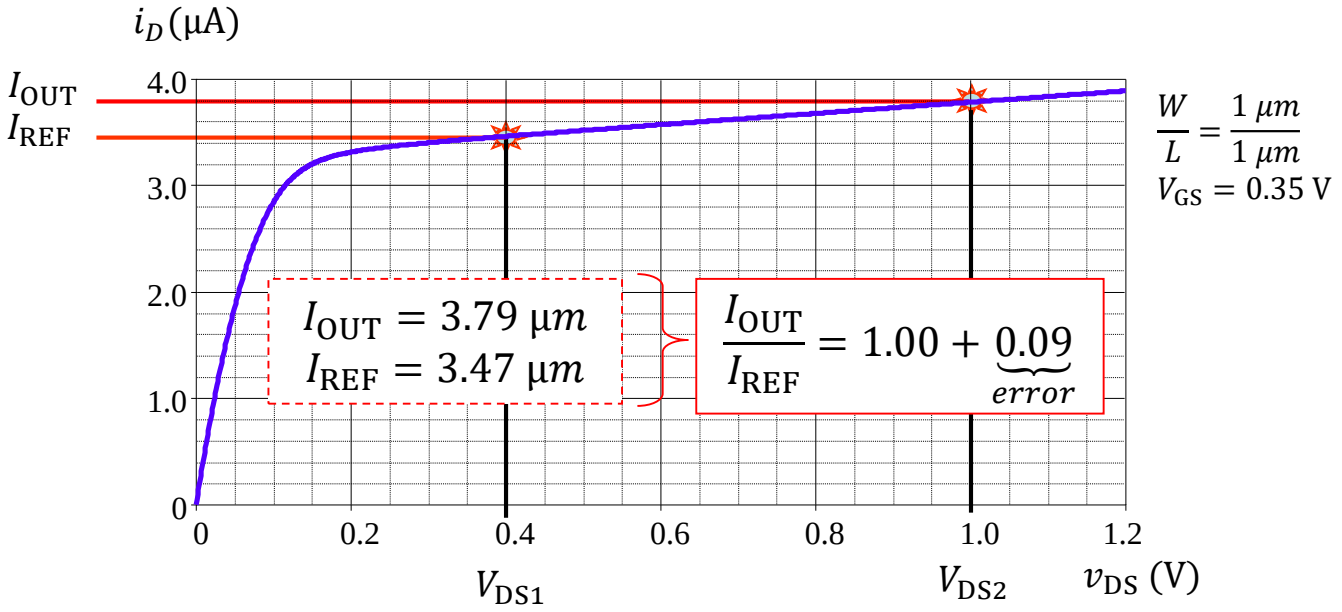
$$i_{OUT} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_2 (V_{GS12} - V_{th})^2(1 + \lambda_2 v_{DS2})$$

channel-length modulation

$$V_{DS1} \neq V_{DS2} \Rightarrow I_{D1} \neq I_{D2}$$

assume: $L_1 = L_2 \Rightarrow \lambda_1 = \lambda_2$



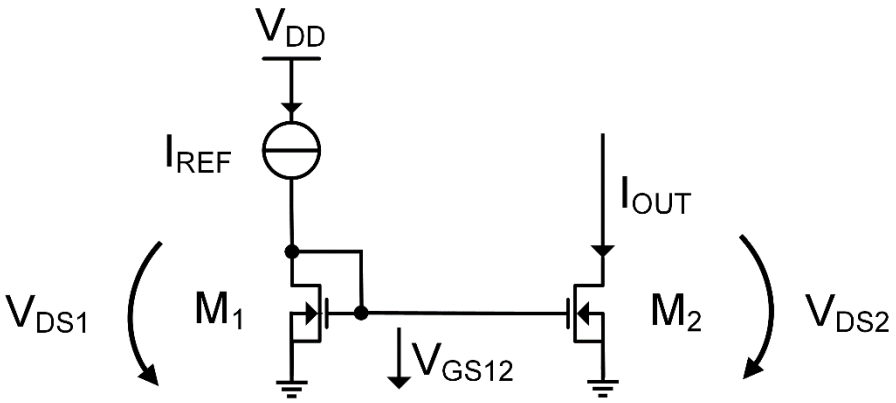



# Impact of Channel-length Modulation

$$I_{OUT} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_2 (V_{GS12} - V_{th})^2(1 + \lambda V_{DS2})$$

$$I_{REF} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_1 (V_{GS12} - V_{th})^2(1 + \lambda V_{DS1})$$

$$\frac{I_{OUT}}{I_{REF}} = \frac{\left(\frac{W}{L}\right)_2 (1 + \lambda V_{DS2})}{\left(\frac{W}{L}\right)_1 (1 + \lambda V_{DS1})}$$




Assume $V_{DS1} \neq V_{DS2}$ and $\left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_2$

$$\frac{I_{OUT}}{I_{REF}} = \frac{(1 + \lambda V_{DS2})}{(1 + \lambda V_{DS1})}_{V_{DS2}=V_{DS1}+\Delta V_{DS}}$$

$$= \frac{1 + \lambda V_{DS1} + \lambda \Delta V_{DS}}{1 + \lambda V_{DS1}} = 1 + \underbrace{\frac{\lambda \Delta V_{DS}}{1 + \lambda V_{DS1}}}_{\text{error term}}$$

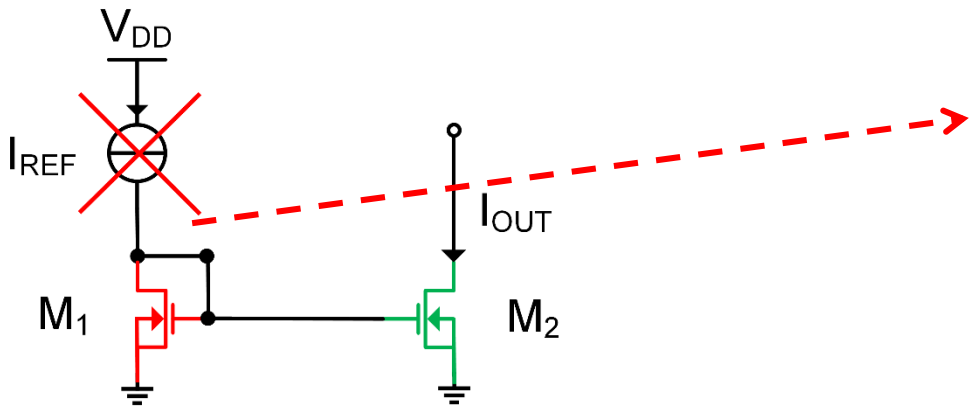
examples: $\Delta V_{DS} = 0.6$ V and $V_{DS1} = 0.35$ V

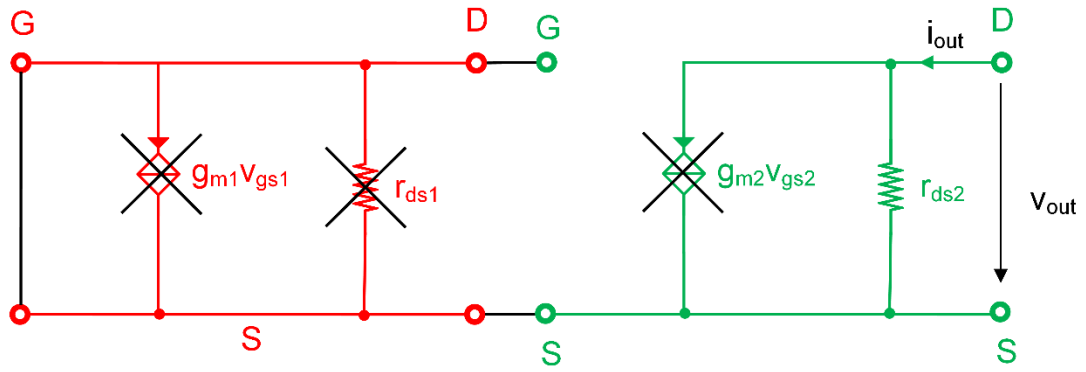
| W/L | λ | error term |
|---|---|---|
| 1.0 µm / 1.0 µm | 0.15 $V^{-1}$ | 0.086 |
| 1.0 µm / 0.2 µm | 0.54 $V^{-1}$ | 0.270 |

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# Small-signal Equivalent Circuit




transistor with gate-drain short
→ "MOS diode"

transistor with constant gate-source voltage

We do not "see" the bias circuit in the small signal equivalent circuit.
It defines the operating point and thus the small-signal parameters, i.e., $r_{\mathrm{ds}}$.

$$r_{\mathrm{out}} = ? = \frac{v_{\mathrm{out}}}{i_{\mathrm{out}}}$$

$$v_{\mathrm{out}} = i_{\mathrm{out}} r_{\mathrm{ds2}}$$

$$\Rightarrow r_{\mathrm{out}} = r_{\mathrm{ds2}}$$

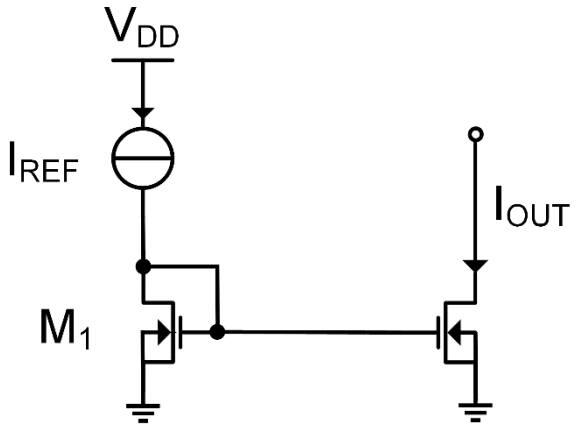
example:

$W/L = 1/1\,, \lambda \approx 0.015\ \mathrm{V}^{-1}$

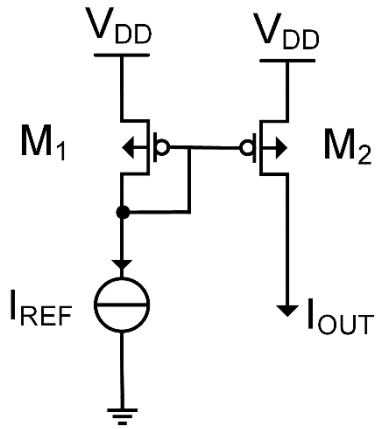
$\mathrm{V_{GS}} = 0.35\ \mathrm{V}, I_{\mathrm{OUT}} \approx 3.7\ \mu\mathrm{A}$

$$r_{\mathrm{out}} \approx \frac{1}{\lambda I_{\mathrm{OUT}}} \approx 18\ \mathrm{M\Omega}$$

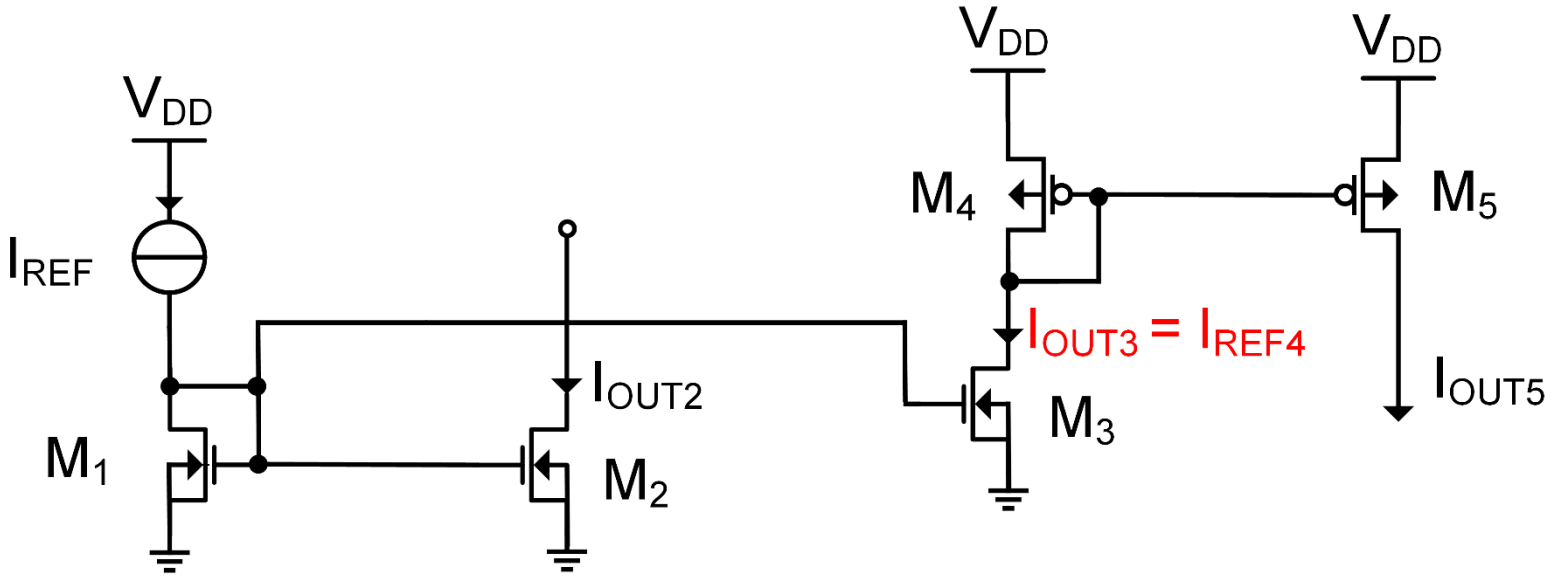
# Current Source & Current Sink




current sink (nMOS)



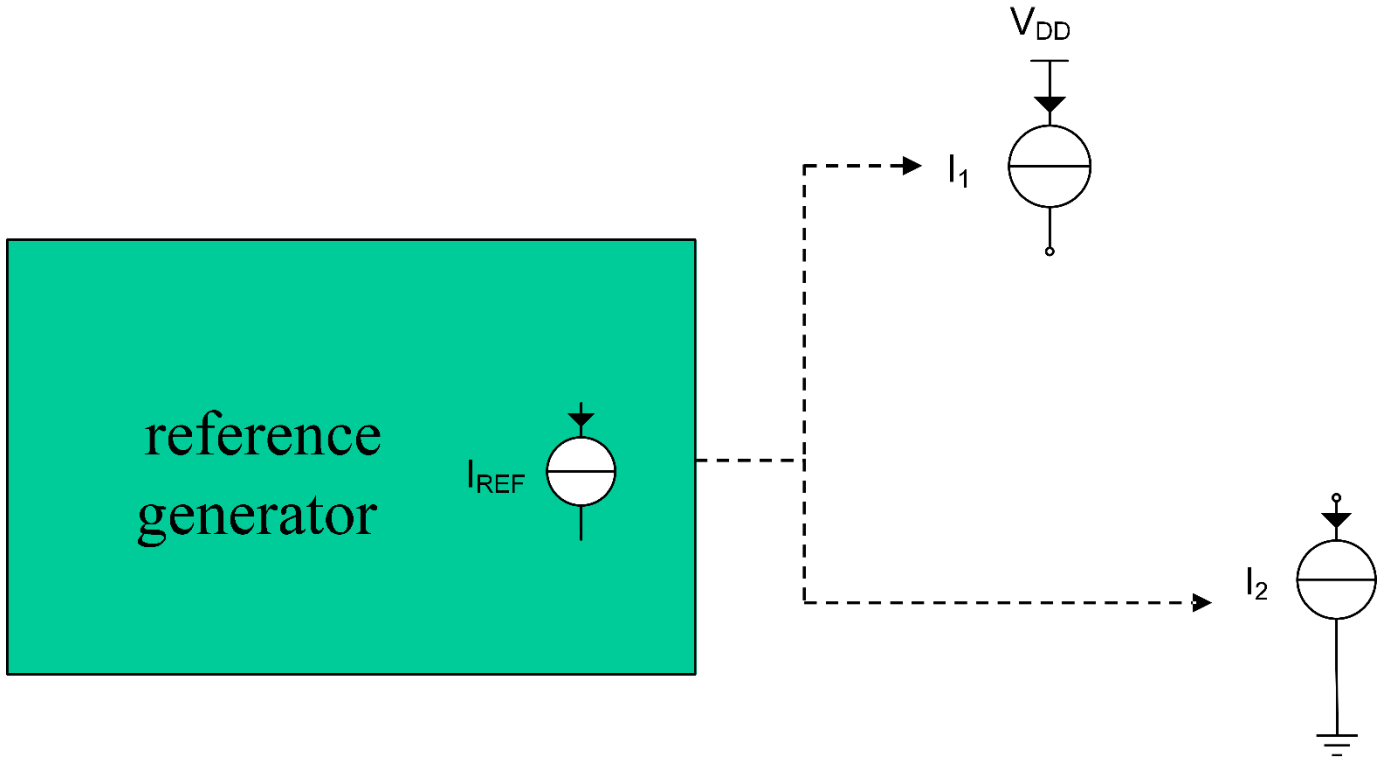

current source (pMOS)



# Biasing – General Approach

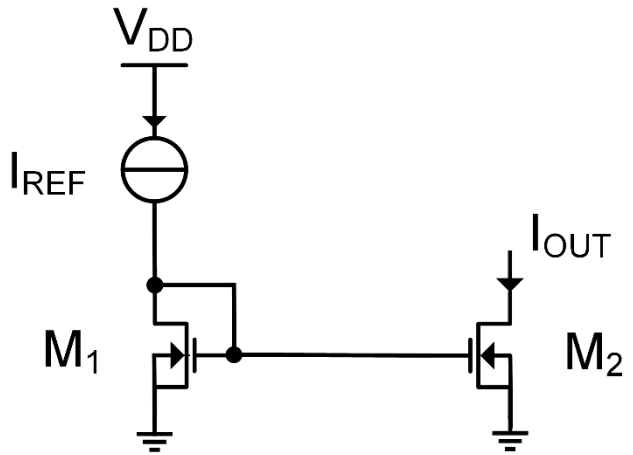
Implement a stable and accurate reference current
(no topic of this lecture).

Copy reference current to circuit blocks that must be biased
by means of current mirrors.



# Summary

- A current mirror generates a (scaled) copy of a reference current.
- Adapt the bias current of the bias branch by the number of diodes.
- Channel-length modulation causes mismatch between the reference and output current
  → use long-channel devices to reduce the mismatch.
- The small-signal output resistance of a current mirror equals the $r_{ds}$ of the output transistor.
- Using devices of the same type yields a better robustness against process variations.



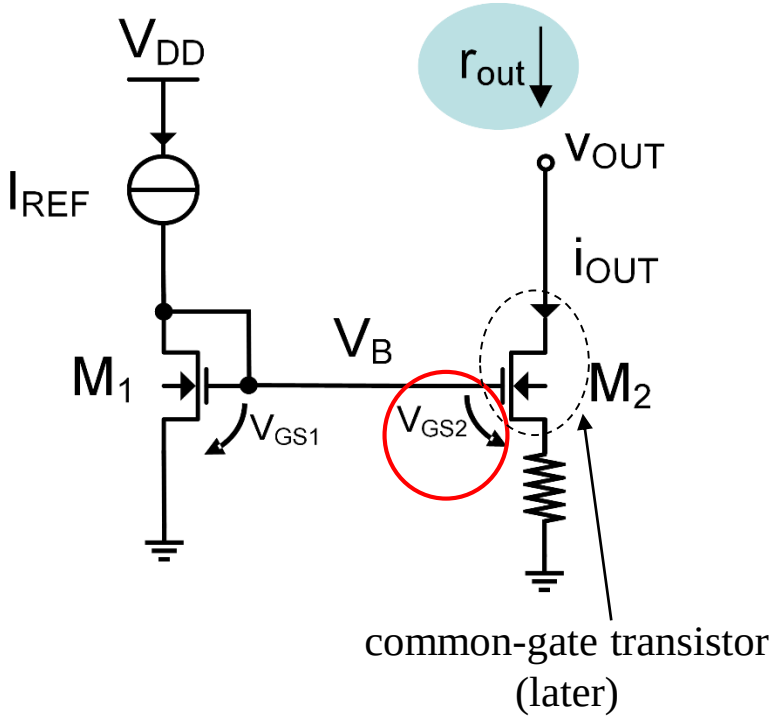
# Literature

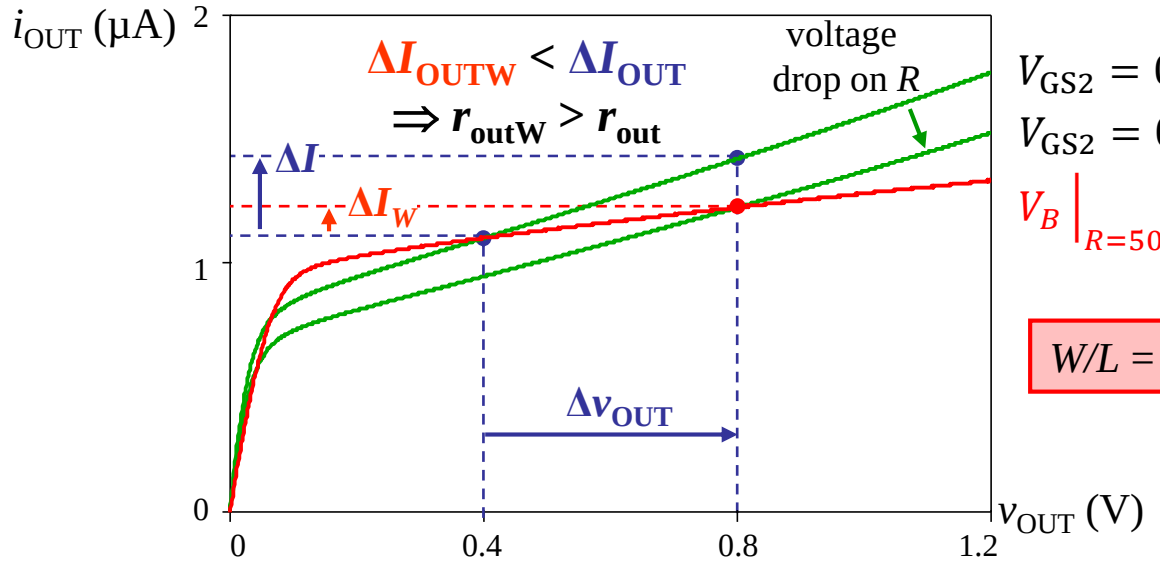
- Literature
- Razavi, chapter 5.1, page 135ff

# Overview

- Amplification – Quick Review
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- Biasing – Current Mirror
- **Cascode Current Source/Sink**

# Widlar Current Source



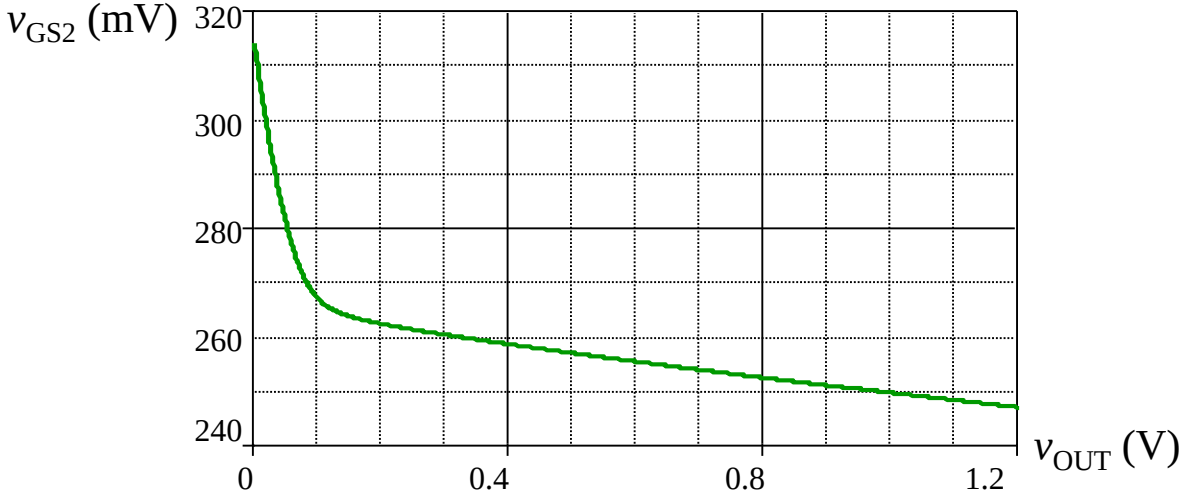




$V_{GS2} = 0.257$ V

$V_{GS2} = 0.251$ V

$V_B\big|_{R=50\,k\Omega} = 0.314$ V

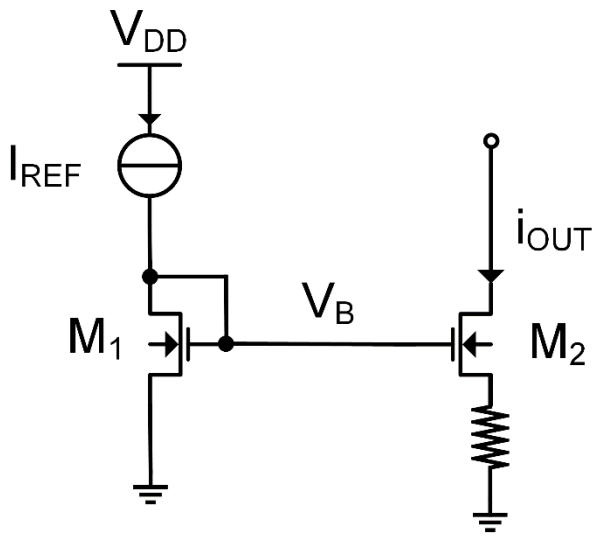
$W/L$ = 2 µm / 0.2 µm

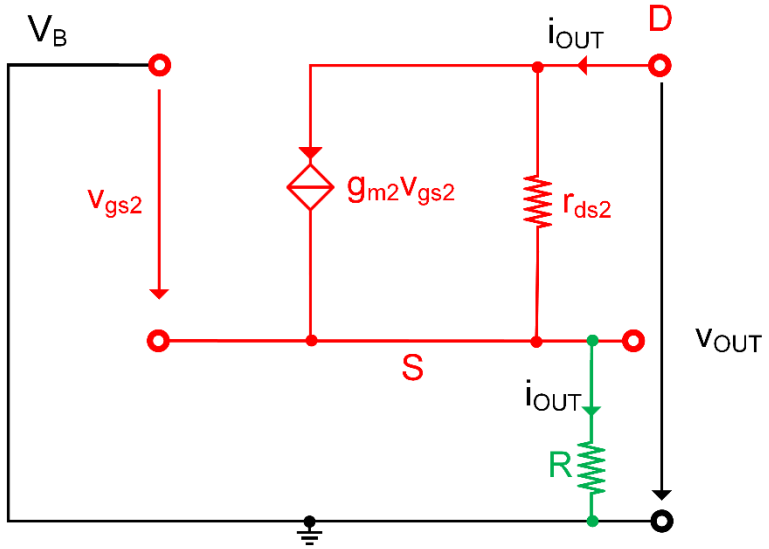



Higher output resistance due to feedback!

This is <u>not</u> a current mirror since $V_{GS1} \neq V_{GS2}$!

# Widlar Current Source – Small-signal Analysis




$$r_{\text{out}} = ? = \frac{v_{\text{out}}}{i_{\text{out}}}$$

$$v_{\text{out}} = (i_{\text{out}} - g_{\text{m2}} v_{\text{gs2}}) r_{\text{ds2}} + R i_{\text{out}}$$

$$v_{\text{gs2}} + i_{\text{out}} R = 0$$

$$r_{\text{out}} = \frac{v_{\text{out}}}{i_{\text{out}}} = (1 + g_{\text{m2}} R) r_{\text{ds2}} + R$$

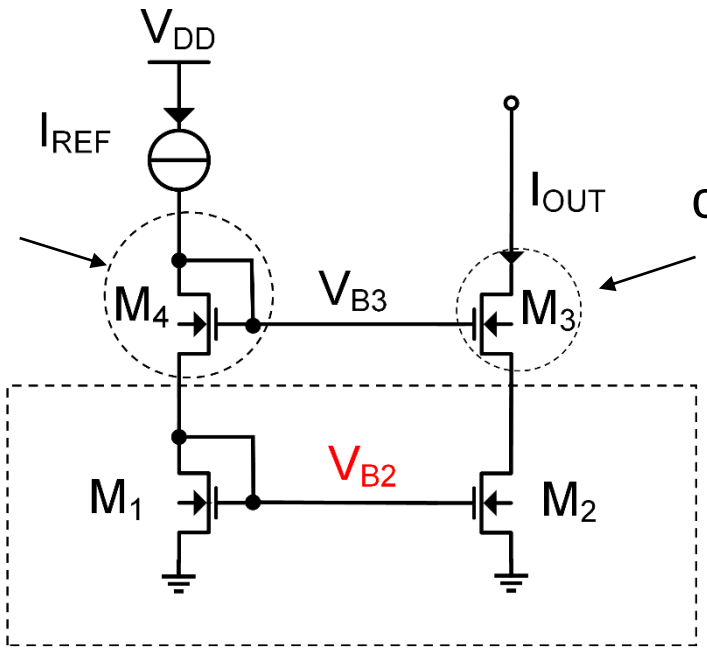
$$= r_{\text{ds2}} + (\cancel{1} + g_{\text{m2}} r_{\text{ds2}}) R \approx r_{\text{ds2}} + g_{\text{m2}} r_{\text{ds2}} R$$

example: $W / L$ = 2µm / 0.2µm, $V_{\text{OUT}}$ = 0.4 V

$$\left.\begin{aligned} r_{\text{ds2}} &= 1.2\ \text{M}\Omega \\ g_{\text{m2}} &= 27.5\ \mu\text{S} \\ R &= 50\ \text{k}\Omega \end{aligned}\right\} \quad r_{\text{out}} \approx 2.9\ \text{M}\Omega \qquad (g_{\text{m2}} r_{\text{ds2}} = 33)$$

Can we implement $R$ by means of a transistor?

# Cascode Current Mirror




generation bias voltage

common-gate transistor

current mirror

Increasing the output resistance is equivalent to decreasing the impact of channel-length modulation.

$$\frac{I_{\mathrm{OUT}}}{I_{\mathrm{REF}}} = \frac{\left(\frac{W}{L}\right)_2 (1 + \lambda V_{\mathrm{DS2}})}{\left(\frac{W}{L}\right)_1 (1 + \lambda V_{\mathrm{DS1}})}$$

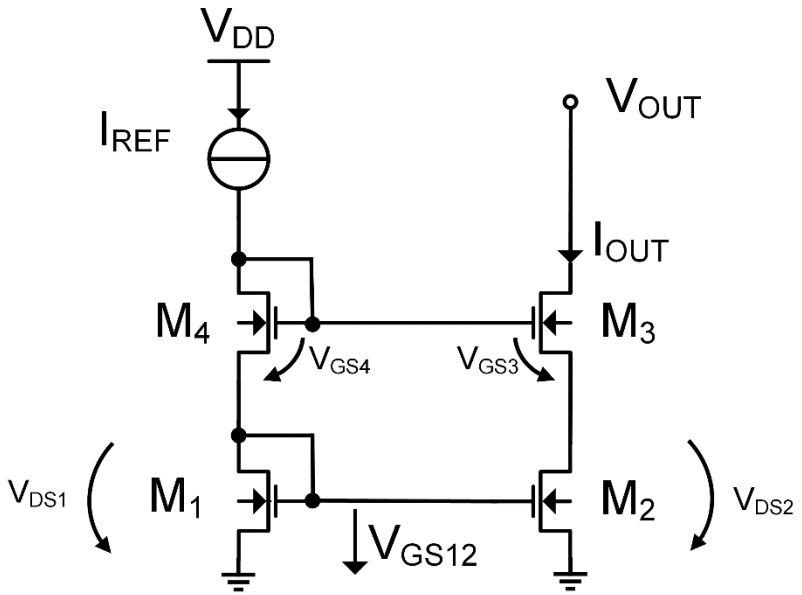
$$r_{\mathrm{out}} \approx r_{\mathrm{ds3}} + g_{\mathrm{m3}} r_{\mathrm{ds3}} R = r_{\mathrm{ds3}} + g_{\mathrm{m3}} r_{\mathrm{ds3}} r_{\mathrm{ds2}} \approx g_{\mathrm{m3}} r_{\mathrm{ds3}} r_{\mathrm{ds2}}$$

example: $r_{\mathrm{out}} = 27.5\ \mu\mathrm{S}\ 1.2\ \mathrm{M\Omega}\ 1.2\ \mathrm{M\Omega} \approx 40\ \mathrm{M\Omega}$ — $W/L = 2\mu m/0.2\mu m$

example: $r_{\mathrm{out}} = 58\ \mu\mathrm{S}\ 8\ \mathrm{M\Omega}\ 8\ \mathrm{M\Omega} \approx 3.7\ \mathrm{G\Omega}$ — $W/L = 10\mu m/10\mu m$

Factor $g_{\mathrm{m3}} r_{\mathrm{DS3}}$ = 460 higher than $r_{\mathrm{out}} = 8\ \mathrm{M\Omega}$ of basic current mirror

# Cascode Current Mirror – Transistor Sizing




current mirror: $\left(\frac{W}{L}\right)_2 / \left(\frac{W}{L}\right)_1$

But what about $T_3$ and $T_4$?

$$V_{B3} = V_{GS4} + V_X = V_{GS3} + V_y$$

objective: $V_{GS4} = V_{GS3}$

$$\Rightarrow V_X = V_Y$$
$$\Leftrightarrow V_{DS1} = V_{DS2}$$

⇒ No impact of channel-length modulation!

$$I_{OUT} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_2 V_{GS12eff}^2 = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_3 V_{GS3eff}^2$$

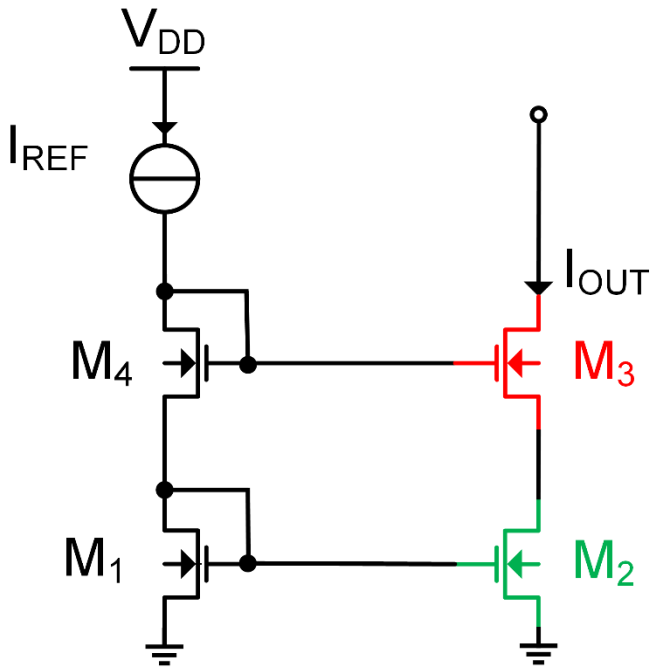
$$I_{REF} = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_1 V_{GS12eff}^2 = \frac{\beta_0}{2}\left(\frac{W}{L}\right)_4 V_{GS4eff}^2$$

(assuming that all transistors have same $\beta_0$ and $V_{th}$)

$$V_{GS3eff} = V_{GS4eff} \rightarrow \frac{\left(\frac{W}{L}\right)_3}{\left(\frac{W}{L}\right)_4} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}$$

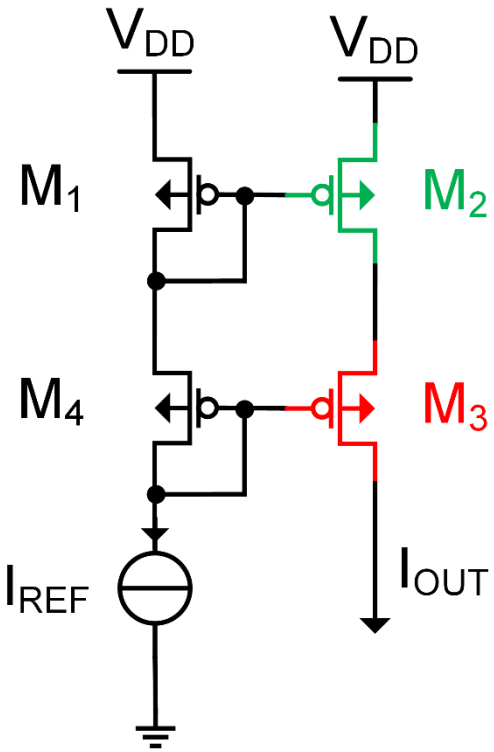
# Cascode Current Source & Cascode Current Sink

$$r_{out} \approx \underbrace{g_{m3} r_{ds3}}_{\gg 1} r_{ds2}$$



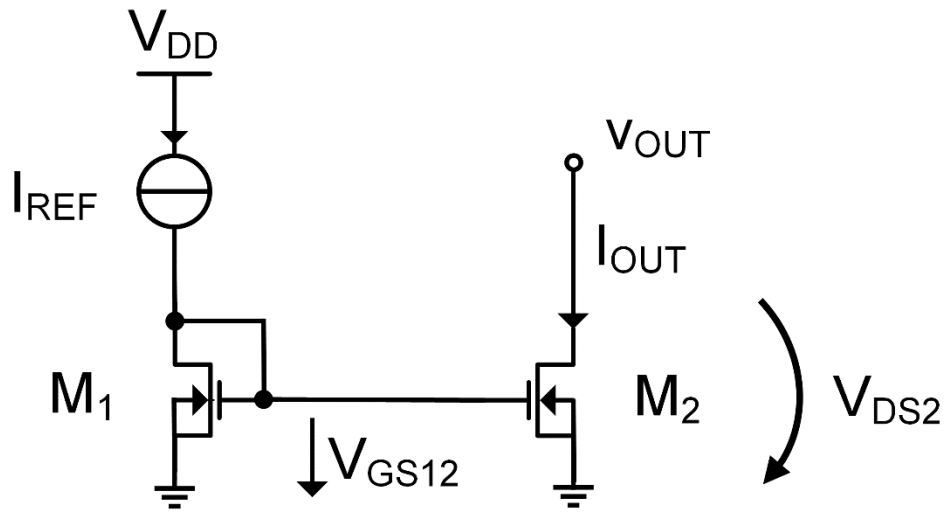

cascode current sink
(nMOS)

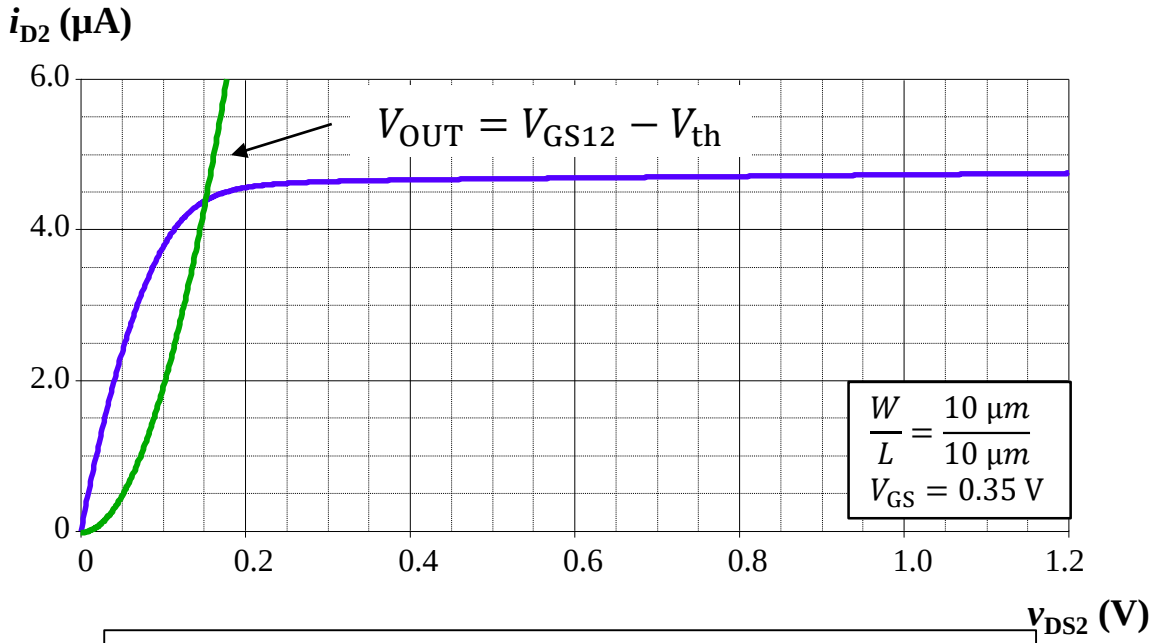
cascode current source
(pMOS)




$$r_{out} \approx \underbrace{g_{m3} r_{ds3}}_{\gg 1} r_{ds2}$$

# Current Mirror – Output Swing




$M_2$ in saturation if

$$v_{DS2} \geq V_{GS12} - V_{th} = V_{GS2eff} = V_{DS2sat}$$

$$\Rightarrow v_{OUT} \geq V_{DS2sat}$$

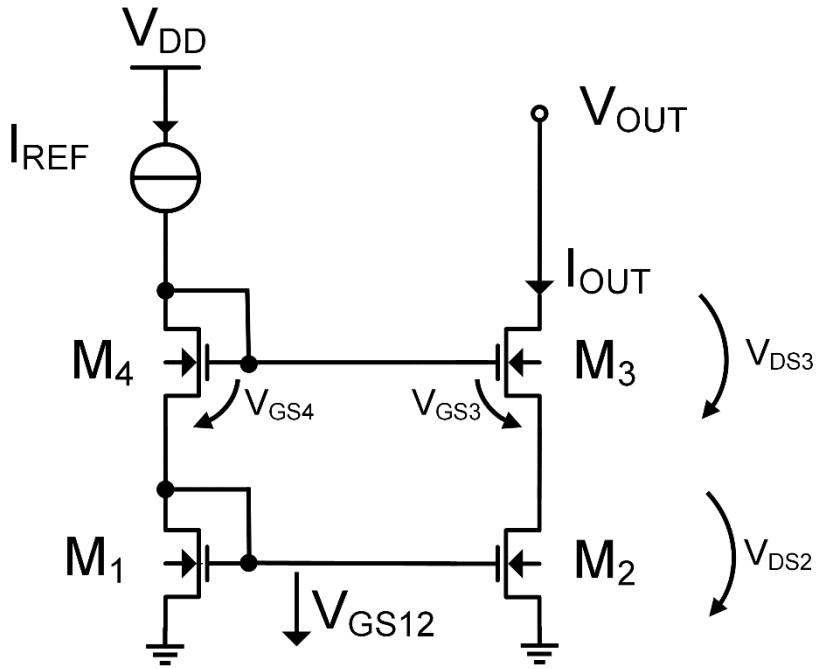
$$I_D = \frac{\beta_0}{2}\frac{W}{L}(V_{GS} - V_{th})$$

$$\frac{W}{L}\uparrow \Rightarrow V_{GSeff}\downarrow$$

$$V_{GS12eff} = V_{DS12sat} = \sqrt{\frac{2I_{REF}}{\beta}} = \sqrt{\frac{2I_{REF}}{\beta_0\left(\frac{W}{L}\right)_1}}$$

Large output swing,
but current not well defined if $v_{DS2} \neq V_{DS1}$

# Cascode Current Source – Output Swing




Saturation of $M_1$ and $M_4$?
Diodes!

M3 in saturation: $v_{DS3} \geq V_{GS3} - V_{th3}$

$$v_{OUT} = v_{DS3} + v_{DS2}$$

$$v_{OUT} \geq V_{GS3} + v_{DS2} - V_{th3}$$

$$V_{GS3} + v_{DS2} = V_{GS4} + V_{GS12}$$

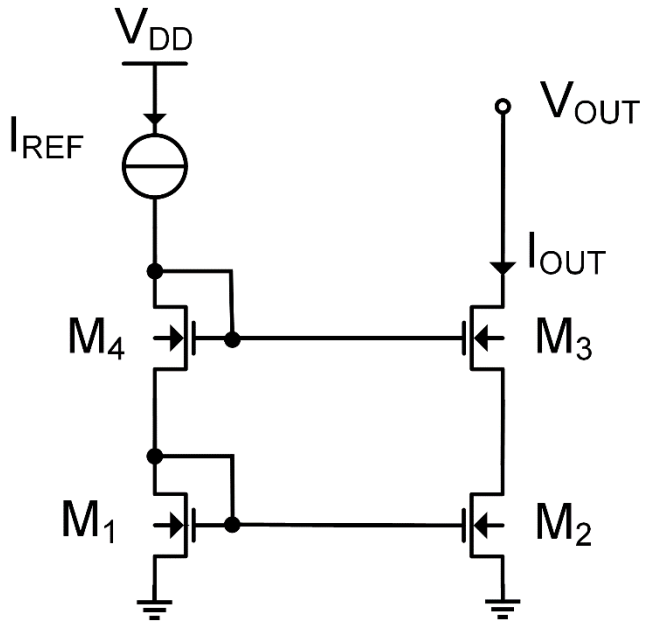
$$v_{OUT} \geq V_{GS12} + V_{GS4} - V_{th3}$$

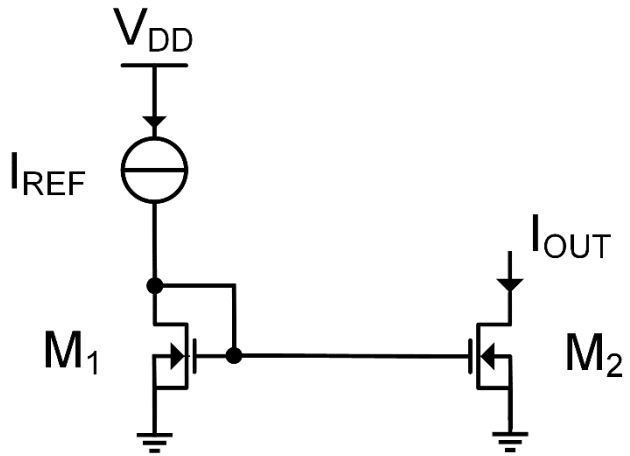
$$\geq V_{GS12eff} + V_{th1} + V_{GS4eff} + \cancel{V_{th4}} - \cancel{V_{th3}}$$

$$v_{OUT} \geq V_{GS12eff} + V_{th1} + V_{GS4eff}$$

# Summary

- The higher the output resistance of a current mirror, the smaller the variations of its output current due to voltage variations.
- A Widlar current source increases its output resistance by means of feedback.
- A cascode current source has a very high output resistance.
- A cascode current source trades output swing for output resistance.



# Literature

- Literature
- Razavi, chapter 5.2, page 139ff